

Exercise: Arrays

1. Read and Display Array Elements

Write a Java program to read five integers into an array and display all the elements in their original order.

```
Test Data
Input 5 integers: 12 7 25 4 18
Expected Output:
The array elements are:
12 7 25 4 18
```

2. Find the Sum and Average of Array Elements

Write a Java program to calculate the sum and average of the elements in an integer array.

```
Test Data
Array elements: 10 20 30 40 50
Expected Output:
Sum: 150
Average: 30.0
```

3. Find the Largest and Smallest Array Elements

Write a Java program to find the largest and smallest values in an integer array.

```
Test Data
Array elements: 45 12 78 3 56
Expected Output:
Largest element: 78
Smallest element: 3
```

4. Search for an Element in an Array

Write a Java program to search for a given integer in an array and display its index. Display a suitable message if the value is not found.

```
Test Data
Array elements: 8 15 22 31 47
Search value: 22
Expected Output:
22 was found at index 2
```

5. Reverse an Array

Write a Java program to display the elements of an integer array in reverse order.

```
Test Data
Array elements: 1 2 3 4 5
Expected Output:
The reversed array is:
5 4 3 2 1
```